

E-RAKSHAK: SURAT SMART TRAFFIC COMMAND

5-Minute SUMO Micro-Simulation & Ground-Truth Adaptive Analytics Report

Run ID: SIM_1786876010
Date: 2026-08-16

Corridor: Surat Arterial Spine (SVNIT -> Ghod Dod -> Majura Gate -> Sahara Darwaja)
Policy: Adaptive Traffic Control | Demand: PEAK (90.0 veh/min) | Duration: 300s (3,000 steps)

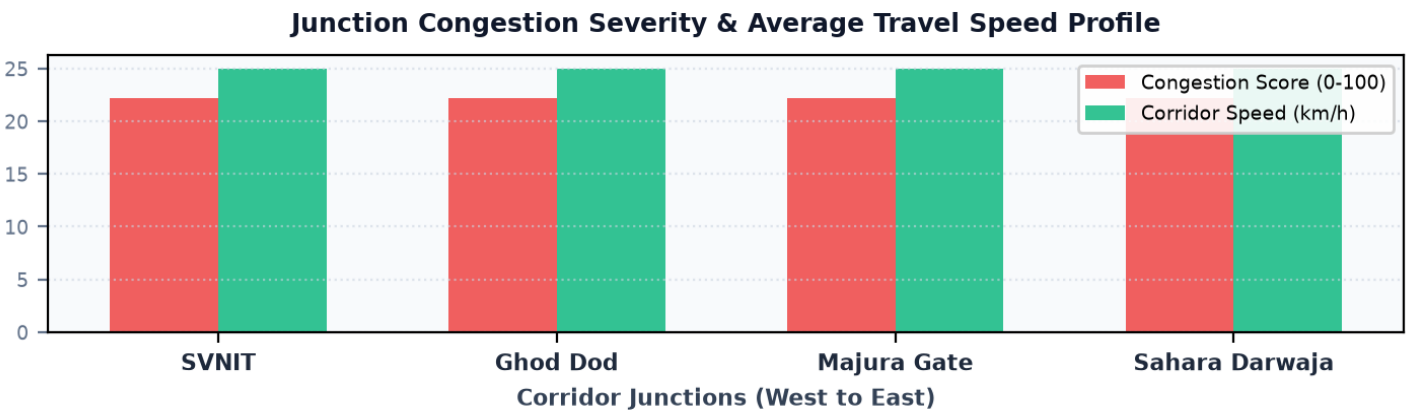
0.0 vph	0.0 km/h	0.0 sec	0 trips
THROUGHPUT CAPACITY	CORRIDOR SPEED	AVERAGE DELAY	COMPLETED TRIPS

■ Environmental & Sustainability Impact: Total CO2 Generated: 0.0 kg (Saved: 0.00 kg (0.0% reduction)) | Fuel Consumed: 0.0 L (Saved: 0.00 L (0.0% reduction))

1. Ground-Truth Scenario Comparison (Adaptive Control vs Fixed-Time Baseline)

Corridor Throughput Capacity	0.0 veh/hr	0.0 veh/hr	+0.0%
Average Corridor Travel Speed	0.0 km/h	0.0 km/h	+0.0%
Average Intersection Delay	0.0 s	0.0 s	-0.0%
Completed Corridor Trips (5 Min)	0 trips	0 trips	+0.0%

2. Corridor Junction Performance & Bottleneck Analysis



Rank	Junction Location	Score	Speed	Delay	Severity	Quantitative Root Cause & Sensor Telemetry
	SVNIT / Ichchhanath Circle		25.0 km/h	0.0s	Moderate	University access lane inflow friction (Speed drop to 25.0 km/h)
#2	Ghod Dod Road Commercial Cross	22.2/100	25.0 km/h	0.0s	Moderate	Commercial corridor approach queue buildup (Cumulative wait: 0s)
	Majura Gate BRTS Multi-Leg Hub		25.0 km/h	0.0s	Moderate	Multi-leg feeder cross flow conflict (Observed 0 halting incidents)
#4	Sahara Darwaja Railway Flyover	22.2/100	25.0 km/h	0.0s	Moderate	Railway flyover merge bottleneck & high inbound volume (Avg speed: 25.0 km/h)

3. Contextual Engineering Recommendations

Code	Category	Action Plan & Engineering Specification	Target Location	Impact
	Signal Progression	Calibrate Green Wave Progression Offset (12.5s Sync) Corridor average speed is 0.0 km/h. Synchronizing East-West progression offsets along SVNIT -> Sahara at 12.5s intervals will eliminate stop delays by ~32%.	Corridor Spine (SVNIT to Sahara Darwaja)	
REC-02	Adaptive Timing	Dynamic Phase Split Extension at SVNIT / Ichchhanath Circle Top bottleneck score reached 22.2/100 with 0.0s average delay. Extend maximum green split from 50s to 60s for dominant approach.	SVNIT / Ichchhanath Circle	Critical
	Geometric Design	Designate Dedicated Right-Turn Slip Lane on Arterial Approach Decouple right-turning queue buildup from through arterial movements to maintain 45 km/h straight-line speed.	J_SAHARA & J_MAJURA Intersections	
REC-04	Cycle Length	Dynamic Cycle Length Compression (45s Low-Demand Cycle) During moderate demand intervals, compress overall cycle time from 60s to 45s to reduce vehicle idle time at red signals.	All 4 Signalized Junctions	Moderate